

Human–AI Co-Agency

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Introduction

Purpose of This Book

Why agency is the central question in AI-supported work

As artificial intelligence becomes increasingly embedded in everyday work, discussions often focus on what AI can do: how powerful systems have become, how quickly they are improving, and how widely they can be deployed. These discussions are not unimportant. Capability, performance, and scale all matter. However, they frequently obscure a more fundamental question—one that determines whether AI use strengthens or undermines human systems over time.

That question is **agency**.

At its core, agency concerns who decides, who is responsible, and who remains accountable when AI systems are introduced into human activity. In AI-supported work, agency is not a philosophical abstraction; it is a practical condition that shapes decision-making, professional judgement, ethical responsibility, and institutional trust. When agency is clear and well supported, AI can augment human capability. When it is ambiguous or eroded, even technically impressive systems can produce fragile, risky, or harmful outcomes.

This book starts from the premise that **human–AI co-agency is the central capability challenge of AI adoption**. Questions of ethics, governance, innovation, and impact all ultimately depend on how agency is distributed and exercised. Without clarity about who retains authority, how decisions are made, and where responsibility lies, other capability efforts struggle to take root.

In many organisations, agency is treated implicitly rather than intentionally. AI tools are introduced to improve efficiency or insight, but little attention is paid to how these tools reshape decision boundaries, influence judgement, or redistribute responsibility. Over time, this leads to subtle but significant shifts: decisions feel less owned, accountability becomes diffuse, and individuals may struggle to articulate their role in AI-mediated processes.

This book exists to make those dynamics visible.

The purpose of *Human–AI Co-Agency* is to clarify what co-agency means in practice, why it is essential for responsible and effective AI use, and how it can be deliberately developed rather than assumed. Rather than offering technical prescriptions or abstract ethical theory, the book focuses on the lived realities of AI-supported work: how people interpret outputs, how authority is exercised, how responsibility is shared, and how judgement is sustained over time.

Co-agency, as used here, does not imply equality between humans and machines, nor does it suggest that AI systems possess moral or institutional agency. Instead, it refers to the **structured, accountable interaction between human actors and AI systems within decision-making processes**. It is about ensuring that humans remain meaningfully in control—not by manually intervening at every step, but by designing roles, boundaries, and escalation paths that preserve agency where it matters most.

This focus distinguishes co-agency from more familiar framings such as “human-in-the-loop” or “AI assistance.” Those concepts often reduce agency to a technical safeguard rather than a socio-organisational capability. Co-agency, by contrast, recognises that agency is shaped by culture, governance, incentives, and professional norms as much as by system design.

Within the AI Capability Framework, co-agency is treated as a **core domain** because it underpins all others. Ethical principles cannot be enacted without clear agency. Governance mechanisms cannot function without defined responsibility. Applied practice becomes risky or incoherent when agency is ambiguous. Reflection and learning break down when it is unclear who is accountable for outcomes.

This book therefore aims to provide readers with a robust, practical understanding of co-agency as a capability that can be cultivated, assessed, and renewed over time. By the end of the book, readers should be able to recognise healthy and unhealthy patterns of co-agency in their own contexts, articulate where agency currently sits, and identify concrete steps for strengthening it.

The Problem of Automation Without Agency

Risks of delegation, abdication, and over-trust

The need for a co-agency perspective becomes clear when we examine a recurring pattern in AI adoption: **automation without agency**.

Automation without agency occurs when AI systems are introduced into decision-making processes without sufficient attention to how responsibility, authority, and judgement are redistributed. In such cases, work may become faster or more scalable, but it also becomes less transparent and less accountable. Over time, this erodes both individual confidence and institutional trust.

This problem rarely arises from malicious intent. More often, it emerges from reasonable but incomplete assumptions. Organisations adopt AI to manage complexity, reduce workload, or improve consistency. Individuals use AI tools to support analysis, drafting, or pattern recognition. In doing so, they may gradually rely on system outputs without revisiting the question of who ultimately owns the decision.

Three related risks tend to characterise automation without agency: **delegation**, **abdication**, and **over-trust**.

Delegation occurs when tasks or decisions are formally or informally handed over to AI systems without clear criteria or limits. In some cases, delegation is explicit: an AI system is authorised to make or recommend decisions within a defined scope. In others, it is implicit: repeated reliance on system outputs gradually shifts expectations about who decides.

Delegation becomes problematic when it is not accompanied by clear accountability. Individuals may follow AI recommendations because they are efficient, consistent, or endorsed, even when those recommendations conflict with contextual knowledge or professional judgement. Over time, the act of deciding becomes conflated with the act of accepting system output.

Abdication is a more subtle but equally concerning risk. It occurs when humans retain nominal responsibility for decisions but lack the practical means or confidence to exercise judgement. In these situations, individuals may feel unable to challenge AI outputs, either because they do not understand how they were produced, because organisational norms discourage dissent, or because responsibility is framed in ways that make challenge seem futile.

Abdication is often reinforced by organisational pressures. Performance metrics, time constraints, or risk-averse cultures can make it easier to defer to AI than to interrogate it. When errors occur, responsibility may be retrospectively assigned to individuals who lacked genuine agency at the time of decision-making.

Over-trust refers to misplaced confidence in AI systems' reliability, objectivity, or appropriateness. This is not simply a matter of misunderstanding technical limitations. Over-trust is shaped by narratives about AI's authority, by the perceived legitimacy of system outputs, and by the social dynamics of decision-making environments.

When over-trust takes hold, AI outputs are treated as neutral or definitive rather than as inputs requiring interpretation. This can lead to reduced scrutiny, normalisation of error, and difficulty recognising drift or bias. Over-trust is particularly dangerous in high-stakes or sensitive contexts, where small errors can have significant consequences.

Together, these risks create a situation in which AI appears to be supporting decision-making while quietly undermining agency. Decisions are made, but ownership is unclear. Responsibility exists, but it is difficult to locate. Learning is hindered because it is unclear who should reflect on outcomes or initiate change.

Automation without agency also has long-term organisational consequences. As reliance on AI increases, individuals may lose confidence in their own judgement or become reluctant to take responsibility for outcomes. New staff may learn to defer rather than to reason. Governance mechanisms may struggle to intervene because practices have become normalised without documentation or review.

Crucially, these dynamics do not require fully autonomous systems to emerge. They can arise in everyday uses of generative AI, decision-support tools, or analytics platforms. The risk is not autonomy itself, but **the gradual erosion of human agency through unexamined reliance**.

This is why co-agency must be treated as a capability rather than a safeguard. Technical controls alone cannot address delegation, abdication, or over-trust. What is required is a deliberate approach to designing roles, responsibilities, and decision processes that preserve meaningful human involvement while allowing AI to contribute productively.

The rest of this book explores how that can be achieved. It defines co-agency in practical terms, examines how it develops over time, identifies common failure modes, and shows how co-agency interacts with ethics, governance, and applied practice. Throughout, the focus remains on a single guiding question:

How can humans work with AI in ways that preserve agency, accountability, and learning—especially as systems become more capable and pervasive?

That question frames the chapters that follow and anchors co-agency as a foundational element of sustainable AI capability.

Position Within the AI Capability Framework

Co-Agency as a Core Capability

How shared agency underpins ethical and effective AI use

Within the AI Capability Framework, **Human–AI Co-Agency** occupies a distinctive position. It is not simply one domain among others, nor is it a specialised concern relevant only to certain contexts or roles. Instead, co-agency functions as a *structural capability*—one that shapes how all other domains are enacted in practice.

At its simplest, co-agency concerns how responsibility, authority, and judgement are distributed when AI systems are involved in human work. At a deeper level, it determines whether AI use strengthens or weakens human decision-making over time. For this reason, co-agency underpins ethical integrity, governance effectiveness, and the sustainability of applied practice.

The AI Capability Framework treats co-agency as a core domain because **AI systems do not act in isolation**. They operate within human systems—organisations, institutions, professions, and social arrangements that assign responsibility, define authority, and create accountability. When AI is introduced into these systems, it inevitably reshapes how decisions are made, how responsibility is perceived, and how outcomes are justified.

Without explicit attention to co-agency, these shifts tend to occur implicitly. Authority may migrate toward system outputs without deliberate choice. Responsibility may become diffuse or ambiguous. Judgement may be exercised less frequently or less confidently. Over time, this can hollow out the very capabilities that AI was intended to support.

Treating co-agency as a capability reframes these dynamics. It recognises that **agency must be actively designed, supported, and renewed**. Co-agency is not guaranteed by keeping a human “in the loop,” nor by assigning nominal responsibility after the fact. It requires clarity about decision boundaries, escalation paths, and interpretive authority throughout AI-supported workflows.

From a capability perspective, ethical and effective AI use is impossible without co-agency. Ethical principles depend on someone being able to exercise judgement and take responsibility for consequences. Governance mechanisms depend on knowing who decides, who reviews, and who intervenes. Applied practice depends on individuals feeling authorised to interpret, question, and adapt AI outputs rather than simply execute them.

In this sense, co-agency acts as a *keystone domain*. When it is strong, other capabilities can function coherently. When it is weak, even well-designed ethical frameworks, governance structures, or technical practices struggle to operate as intended.

Importantly, co-agency is not a fixed arrangement. It evolves as technologies, organisational contexts, and forms of work change. Early experimentation with AI often involves informal co-agency, where roles and responsibilities are negotiated tacitly. As use becomes more embedded, co-agency must be stabilised through clearer structures and shared understanding. At more advanced stages, co-agency becomes adaptive, allowing humans to recalibrate decision boundaries as systems and contexts evolve.

By positioning co-agency as a core capability, the framework emphasises that **agency is not something that can be “left to culture” or “assumed to persist”**. It must be made visible, discussed explicitly, and revisited over time. This book exists to support that work.

Relationships to Other Domains

Links to ethics, governance, and applied practice

While co-agency is a distinct domain within the AI Capability Framework, it does not operate independently. Its significance lies in how it connects and gives effect to other domains—particularly ethics, governance, and applied practice. Understanding these relationships is essential for avoiding fragmented or superficial capability development.

Co-Agency and Ethics, Equity, and Impact

Ethical AI use depends fundamentally on agency. Ethical principles such as fairness, accountability, transparency, and responsibility are meaningful only if there are humans who can interpret them, apply them, and act on them in specific situations. Without co-agency, ethics risks becoming abstract or symbolic rather than operational.

When co-agency is weak, ethical responsibility often becomes displaced. Individuals may assume that ethical concerns are “handled by the system,” “covered by policy,” or “someone else’s job.” In such environments, harms may go unrecognised or unaddressed not because people are indifferent, but because they lack the authority or clarity needed to intervene.

Strong co-agency enables ethical capability by ensuring that:

- responsibility for ethical judgement is clearly located,
- individuals feel authorised to question AI-supported decisions,
- ethical concerns can be escalated and reviewed,
- and learning from ethical dilemmas informs future practice.

Equity and impact considerations are particularly sensitive to agency. AI systems may differentially affect groups or individuals, often in ways that are not immediately visible. Recognising and responding to these effects requires human interpretation, contextual knowledge, and moral judgement. Co-agency ensures that such interpretation remains part of decision-making rather than being sidelined by efficiency or automation.

In this way, co-agency does not replace ethical reasoning; it **enables it to function in practice**.

Co-Agency and Decision-Making and Governance

Governance structures are only as effective as the agency they support. Policies, oversight bodies, and accountability mechanisms rely on clear understandings of who decides, who reviews, and who is responsible for outcomes. Co-agency provides the connective tissue that allows governance to operate meaningfully.

When co-agency is underdeveloped, governance often becomes procedural rather than purposeful. Rules may exist, but they fail to influence everyday practice. Approval processes may be bypassed or treated as formalities. Accountability may be retrospective and punitive rather than constructive.

By contrast, strong co-agency supports governance by:

- clarifying decision boundaries and escalation routes,
- enabling individuals to act within governance frameworks confidently,
- making responsibility visible at appropriate levels,
- and supporting timely intervention when practices drift.

Co-agency also helps governance adapt. As AI use evolves, rigid governance structures can become misaligned with practice. Co-agency allows organisations to recognise these misalignments early and adjust governance in response, rather than relying solely on compliance or enforcement.

In this sense, co-agency and governance are mutually reinforcing. Governance provides structure and legitimacy; co-agency ensures that structure is inhabited and enacted by human judgement.

Co-Agency and Applied Practice and Innovation

Applied practice is where AI capability becomes visible. It is also where co-agency is most directly tested. Day-to-day use of AI systems involves countless micro-decisions: how to interpret outputs, when to rely on them, when to override them, and when to stop using them altogether.

Without co-agency, applied practice tends to drift toward automation by default. AI outputs become accepted as “the answer,” and human involvement becomes limited to execution or oversight without genuine authority. This may appear efficient in the short term, but it undermines learning, accountability, and adaptability.

Strong co-agency supports applied practice by:

- legitimising human interpretation and challenge,
- enabling experimentation within clear boundaries,
- supporting documentation and reflection,
- and ensuring that innovation remains aligned with organisational purpose and values.

Innovation, in the AI Capability Framework, is not equated with novelty or speed. It is understood as the ability to explore and adapt responsibly. Co-agency is what makes such exploration safe and sustainable. It allows organisations to learn from experimentation without normalising risk or eroding trust.

Applied practice also feeds back into co-agency. As individuals gain experience working with AI systems, their understanding of appropriate roles and boundaries deepens. Reflection on practice informs refinements to co-agency arrangements, creating a virtuous cycle of learning and adaptation.

Co-Agency as a Connector Domain

Taken together, these relationships highlight why co-agency cannot be treated as a specialist concern or a secondary consideration. It connects ethical intent to practical action, governance structures to lived decision-making, and innovation to accountability.

Within the AI Capability Framework, co-agency functions as a **connector domain**. It translates abstract principles into concrete behaviour. It ensures that capability development remains human-centred even as systems become more capable. It provides the conditions under which other domains can operate coherently rather than in isolation.

This book builds on that positioning. The chapters that follow define co-agency more precisely, explore how it develops over time, identify common failure modes, and show how it can be strengthened across different contexts. Throughout, the focus remains on enabling humans to retain meaningful agency in AI-supported work—not by resisting technology, but by shaping how it is integrated into human systems.

Defining Human–AI Co-Agency

What Co-Agency Is

Shared responsibility, decision boundaries, and role clarity

Human–AI co-agency refers to the **structured and accountable way in which humans and AI systems jointly participate in work, decision-making, and sense-making**, while responsibility and authority remain clearly human. It is not a claim about what AI is, but a description of how AI is *situated within human systems*.

At its core, co-agency is about **how decisions happen** when AI systems are involved: who initiates action, who interprets outputs, who decides whether to act, and who is accountable for consequences. These questions are often left implicit in AI adoption, but they determine whether AI use strengthens or weakens human capability over time.

Co-agency recognises that AI systems increasingly shape human activity without acting independently. They generate options, surface patterns, draft content, rank alternatives, and influence attention. These contributions can be powerful, but they do not constitute agency in the human sense. AI systems do not hold values, bear responsibility, or participate in institutional accountability. Co-agency therefore does not mean shared moral or legal agency; it means **designed interaction** within human responsibility structures.

Three elements are central to co-agency as a capability: **shared responsibility**, **decision boundaries**, and **role clarity**.

Shared responsibility

Shared responsibility does not mean diluted responsibility. In co-agency, responsibility remains human, but it is often **distributed across roles and levels**. Individuals, teams, and institutions may each carry responsibility for different aspects of AI-supported work: interpretation, approval, oversight, escalation, or learning.

Co-agency requires that these responsibilities are:

- explicit rather than assumed,
- understood by those who hold them,
- and supported by appropriate authority and resources.

When responsibility is unclear, individuals may feel exposed or constrained. They may follow AI outputs defensively to avoid blame, or disengage from judgement altogether. Conversely, when responsibility is well-designed, individuals are more likely to exercise judgement confidently and transparently.

Shared responsibility also includes responsibility for **learning**. Co-agency is not only about who decides, but about who reflects, who revises practice, and who initiates change when AI use produces unexpected outcomes. Capability depends on responsibility extending beyond execution to include interpretation and renewal.

Decision boundaries

Decision boundaries define **where AI contribution ends and human judgement begins**, and where that judgement may be escalated or reviewed. These boundaries are rarely fixed; they vary by context, risk, and maturity. What matters is not rigid specification, but clarity and adaptability.

In capable co-agency arrangements:

- AI outputs are treated as inputs to decision-making, not as decisions themselves.
- Humans understand when reliance on AI is appropriate and when it is not.
- There are recognised points at which judgement must be exercised, questioned, or escalated.

Decision boundaries help prevent both over-automation and under-use. They legitimise challenge and override without framing them as failure. They also support consistency by reducing reliance on individual discretion alone.

Crucially, decision boundaries are not only technical. They are organisational and cultural. Even when systems are designed to allow human intervention, social norms or performance pressures may discourage it. Co-agency therefore requires boundaries to be reinforced through leadership signals, governance structures, and professional norms.

Role clarity

Role clarity concerns how human roles are defined and understood in AI-supported systems. This includes not only formal roles, such as decision-makers or reviewers, but also informal roles related to interpretation, translation, and sense-making.

In many AI deployments, roles are blurred. Individuals may be expected to “use AI responsibly” without guidance on what that responsibility entails. Others may be assigned nominal accountability without access to the information or authority needed to fulfil it. Co-agency addresses this by making roles explicit and aligning them with actual practice.

Role clarity supports:

- confidence in decision-making,
- accountability that feels legitimate rather than imposed,
- and coordination across teams and functions.

It also supports learning. When roles are clear, it is easier to identify where breakdowns occurred and where improvements are needed. When roles are ambiguous, learning is often lost in blame or defensiveness.

Taken together, shared responsibility, decision boundaries, and role clarity define co-agency as a **designed capability** rather than an emergent by-product of AI use. They enable humans to remain meaningfully involved in AI-supported work without reverting to manual control or resisting augmentation.

Importantly, co-agency does not imply symmetry. Humans and AI systems contribute in different ways, with different forms of authority. Co-agency is not about balancing power, but about **preserving human agency within augmented systems**.

What Co-Agency Is Not

Why co-agency is not autonomy, replacement, or partnership rhetoric

Clarifying what co-agency is *not* is essential, because many popular narratives about AI obscure or distort the concept. These narratives often arise from technical, commercial, or rhetorical framings that emphasise capability without responsibility.

Co-agency is not autonomy

Co-agency should not be confused with system autonomy. Autonomous AI systems are often defined by their ability to act without human intervention. Co-agency, by contrast, is defined by **intentional human involvement**.

Autonomy narratives tend to focus on efficiency and scale, framing human intervention as friction. From a capability perspective, this framing is misleading. Human judgement is not a bottleneck to be eliminated; it is a capability to be preserved.

Even when AI systems operate automatically within defined parameters, co-agency requires that humans:

- set those parameters intentionally,
- monitor outcomes,
- and retain authority to intervene, revise, or withdraw use.

Autonomy without agency creates systems that are difficult to govern, difficult to learn from, and difficult to justify ethically. Co-agency insists that meaningful human responsibility remains intact, even as systems become more capable.

Co-agency is not replacement

Co-agency also differs fundamentally from replacement narratives. Claims that AI will replace human roles often focus on task substitution rather than on the broader functions those roles serve. While AI may perform certain tasks more efficiently, it does not replace the **institutional and ethical functions of human roles**.

Replacement framings obscure the ways in which AI reshapes work rather than eliminates it. They encourage abdication by framing human involvement as temporary or obsolete. In practice, this often leads to hidden labour: humans continue to interpret, correct, and contextualise AI outputs without recognition or authority.

Co-agency rejects replacement as a goal. Instead, it asks how human roles must evolve in response to AI, and how responsibility should be redistributed rather than erased. It recognises that many human contributions—such as judgement under uncertainty, ethical reasoning, and accountability—cannot be automated without undermining legitimacy and trust.

Co-agency is not partnership rhetoric

Finally, co-agency should not be confused with rhetorical claims about “AI as a partner” or “collaboration between humans and machines.” While such language may be intended to humanise technology, it often creates conceptual confusion.

Partnership implies shared agency, mutual responsibility, and reciprocal obligation. AI systems do not possess these qualities. Treating them as partners risks obscuring where responsibility truly lies and may encourage misplaced trust or emotional attachment.

Co-agency does not anthropomorphise AI. It treats AI systems as powerful tools and contributors within human systems, not as social actors. This distinction is critical for maintaining accountability and avoiding moral confusion.

By rejecting autonomy, replacement, and partnership rhetoric, co-agency maintains a clear boundary between **capability and responsibility**. It allows organisations to take advantage of AI’s strengths without surrendering control or obscuring accountability.

Co-Agency as a practical framing

What distinguishes co-agency from these alternatives is its **practical orientation**. It is not a vision of the future, but a framework for present-day decision-making. It asks concrete questions:

- Who decides?
- Who is responsible?
- Where are boundaries drawn?
- How is learning supported?

These questions can be answered differently across contexts, but they cannot be ignored. Co-agency provides a way to answer them intentionally rather than by default.

In doing so, it shifts AI adoption from a technical exercise to a capability-building process. It acknowledges that AI will continue to change, but insists that human agency must remain visible, accountable, and renewable.

The next sections of this book build on this definition by examining how co-agency is enacted in practice, how it develops over time, and how common failure modes can be identified and addressed. Together, they show how co-agency can move from an abstract concern to a lived organisational capability.

Co-Agency in Practice

Human Roles in AI-Supported Work

Judgement, interpretation, accountability

Human–AI co-agency becomes real not through policy statements or system architecture, but through **human roles enacted in practice**. Regardless of how advanced AI systems become, it is human judgement, interpretation, and accountability that determine whether AI use strengthens or undermines capability.

In AI-supported work, humans do not simply “use” AI. They **frame problems, interpret outputs, decide whether to act, and take responsibility for consequences**. These roles are often compressed into a single moment of interaction, but analytically they are distinct. Co-agency depends on keeping these functions visible and supported rather than allowing them to collapse into unexamined acceptance of system output.

Judgement

Judgement is the capacity to decide *whether* and *how* to act in a specific context, given incomplete information and competing values. AI systems may generate options, rankings, or recommendations, but they do not exercise judgement in this sense. They do not understand stakes, legitimacy, or institutional responsibility.

In practice, judgement involves:

- assessing the relevance of AI output to the current context,
- recognising when outputs conflict with contextual knowledge or ethical considerations,
- deciding when to rely on AI and when to override or ignore it.

Judgement is particularly critical in situations of uncertainty, novelty, or high stakes. These are precisely the situations in which AI outputs may appear persuasive while being least reliable. Co-agency requires that individuals are not only permitted but **expected** to exercise judgement, even when this slows down decision-making or introduces friction.

When judgement is unsupported, individuals may default to system outputs to avoid risk or responsibility. Over time, this erodes professional confidence and weakens organisational learning. Strong co-agency treats judgement as a capability to be cultivated, not a liability to be minimised.

Interpretation

Interpretation sits between system output and decision. AI systems produce outputs that require meaning-making: probabilities, suggestions, drafts, summaries, or classifications. These outputs do not speak for themselves. Their significance depends on context, intent, and consequence.

Human interpretation involves:

- understanding what an AI output represents and what it does not,
- situating outputs within broader evidence or experience,
- recognising uncertainty, bias, or limitation.

Interpretation is often invisible labour. Individuals may be expected to “sense-check” or “apply judgement” without formal recognition or support. In weak co-agency arrangements, interpretation is treated as an informal skill rather than a defined role, making it difficult to resource, train, or evaluate.

Co-agency strengthens interpretation by:

- acknowledging it as a core human contribution,
- clarifying when interpretation is required,
- and ensuring that interpreters have access to relevant context and authority.

This is particularly important in collaborative environments, where interpretations must be shared, discussed, and sometimes contested. Co-agency legitimises disagreement and deliberation rather than treating divergence from AI output as inefficiency or error.

Accountability

Accountability anchors co-agency in institutional reality. It concerns who is answerable for decisions and outcomes, both internally and externally. While AI systems may influence decisions, they cannot be accountable in organisational, legal, or ethical terms.

In AI-supported work, accountability often becomes blurred. Individuals may feel responsible for outcomes without having exercised real authority, or they may defer to AI outputs while retaining nominal accountability. Both situations are problematic.

Effective co-agency aligns accountability with:

- decision authority,
- access to information,
- and the ability to intervene or escalate.

This alignment ensures that responsibility is neither unfairly imposed nor quietly displaced. It also supports learning. When accountability is clear, organisations can examine outcomes constructively rather than defensively, identifying where judgement, interpretation, or system design needs adjustment.

Accountability in co-agency is not solely about blame or liability. It is also about **ownership of learning**. Who is responsible for reviewing outcomes? Who initiates change when practices drift? Who ensures that lessons are integrated into future work? These questions are central to sustaining capability over time.

Taken together, judgement, interpretation, and accountability define the human side of co-agency. They are not optional add-ons to AI use, but the conditions under which AI can contribute productively to human systems.

AI Roles in Human Systems

Augmentation, pattern detection, and support functions

Understanding co-agency also requires clarity about what AI systems *do* within human systems—and, equally importantly, what they do not do. Co-agency does not assign agency to AI; it assigns **functions** that support human work.

In practice, AI systems contribute in three broad ways: **augmentation**, **pattern detection**, and **support functions**. These roles are descriptive rather than prescriptive; they help clarify how AI can be integrated without obscuring responsibility.

Augmentation

Augmentation refers to AI's ability to extend human capacity by reducing cognitive load, accelerating routine tasks, or enabling exploration at scale. In this role, AI supports humans in doing work they already do, more efficiently or more expansively.

Examples of augmentation include:

- drafting or summarising content for human review,
- generating alternative options or perspectives,
- supporting scenario exploration or brainstorming.

In co-agency terms, augmentation is effective when it:

- leaves decision authority clearly with humans,
- invites interpretation rather than compliance,

- and supports learning rather than substitution.

Problems arise when augmentation quietly becomes delegation. When AI-generated drafts or recommendations are accepted by default, augmentation shifts toward automation. Co-agency requires that augmentation remains explicitly framed as support, not replacement.

Pattern detection

AI systems excel at identifying patterns across large datasets or complex inputs. This capability can be valuable in highlighting trends, anomalies, or correlations that humans may struggle to detect unaided.

In human systems, pattern detection supports:

- situational awareness,
- risk identification,
- and exploratory analysis.

However, patterns are not explanations, and correlations are not judgements. AI systems surface signals; humans must decide what those signals mean and how to respond. Co-agency depends on maintaining this distinction.

Pattern detection becomes risky when:

- outputs are treated as conclusions rather than prompts,
- uncertainty is ignored,
- or contextual factors are excluded.

Effective co-agency ensures that pattern detection feeds into human sense-making processes rather than bypassing them. It also requires awareness of how training data, assumptions, and system design shape what patterns are visible and which are obscured.

Support functions

Beyond augmentation and pattern detection, AI systems increasingly perform support functions that structure human work. These may include organising information, prioritising tasks, or mediating access to resources.

Support functions influence behaviour indirectly. By shaping attention, workflow, or perceived relevance, AI systems can steer decisions without making them explicitly. This makes support functions particularly significant from a co-agency perspective.

Co-agency requires that:

- the influence of support functions is recognised,
- humans retain the ability to override or reconfigure them,
- and their effects are subject to reflection and review.

When support functions operate invisibly, they can subtly erode agency by narrowing choice or normalising certain behaviours. Making these functions visible and contestable is essential for maintaining meaningful human involvement.

AI roles as designed contributions

Across these roles, a key principle of co-agency is that **AI contributions are designed, not discovered**. How AI augments, detects patterns, or supports work depends on choices made by humans: choices about scope, boundaries, and integration.

Treating AI roles as designed contributions reinforces accountability. It reminds organisations that AI behaviour is not inevitable, and that human decisions shape how systems function in practice. This perspective also supports adaptation. As needs and contexts change, AI roles can be reconfigured without abandoning co-agency.

Importantly, recognising AI roles does not require detailed technical understanding. What matters is functional clarity: knowing what AI is contributing, how that contribution influences decisions, and where human judgement remains essential.

Co-Agency as a practical arrangement

Co-agency in practice emerges from the **alignment of human roles and AI roles**. When judgement, interpretation, and accountability are clearly supported, and AI functions are well understood, co-agency becomes a lived capability rather than an abstract ideal.

Breakdowns in co-agency often occur at the seams: where interpretation is assumed but unsupported, where accountability exists without authority, or where AI support quietly becomes directive. Recognising these seams is the first step toward strengthening capability.

The following chapters examine how co-agency develops over time, how risks and failure modes emerge, and how co-agency can be integrated with ethics and governance. Together, they show how co-agency can move from an implicit assumption to an intentional, renewable capability embedded in everyday practice.

Developing Co-Agency Over Time

Human–AI co-agency is not established at the moment AI is introduced, nor is it achieved through a single design decision. It develops over time, shaped by experience, reflection, and organisational learning. As AI use evolves, so too must the ways in which responsibility, judgement, and authority are exercised.

This chapter describes three **developmental patterns** commonly observed as co-agency matures: early co-agency, developing co-agency, and advanced co-agency. These are not stages to be completed in sequence, nor benchmarks to be achieved uniformly. In practice, organisations may exhibit different patterns simultaneously across teams, functions, or contexts.

The purpose of this framing is not classification, but **recognition**. Understanding how co-agency typically develops helps organisations anticipate challenges, avoid common pitfalls, and support capability growth deliberately rather than reactively.

Early Co-Agency

Exploration, uncertainty, and experimentation

Early co-agency emerges when AI systems are first introduced into a context. At this stage, use is often exploratory, informal, and unevenly distributed. Individuals and teams experiment with AI to understand what it can do, where it might be useful, and how it fits into existing work.

This phase is characterised by **uncertainty**. Roles and responsibilities are rarely explicit. Decision boundaries are fluid or unarticulated. Individuals rely on personal judgement and informal norms to decide when and how to use AI. This uncertainty is not inherently problematic; in fact, it is often necessary for learning.

In early co-agency, experimentation plays a valuable role. It allows people to test assumptions, surface limitations, and discover opportunities that were not apparent in abstract planning. It also helps organisations identify where AI genuinely adds value and where it creates friction or risk.

However, early co-agency is also fragile. Because roles and boundaries are unclear, responsibility may be unevenly distributed. Some individuals may take on disproportionate risk, while others remain disengaged. Judgement may be exercised inconsistently, and learning may remain local rather than shared.

Common characteristics of early co-agency include:

- reliance on individual initiative rather than collective practice,
- informal interpretation of AI outputs,
- limited documentation or review,
- and uncertainty about accountability when outcomes are contested.

At this stage, failures or missteps are often treated as personal rather than systemic. This can discourage transparency and learning if individuals feel exposed or unsupported.

Capability-oriented organisations recognise early co-agency as a **learning phase**, not a compliance problem. They focus on creating psychological safety, encouraging

reflection, and capturing insight rather than enforcing premature standardisation. At the same time, they remain attentive to risks, particularly where experimentation intersects with high-stakes decisions.

The transition out of early co-agency begins when organisations recognise recurring patterns in use and outcome, and when the need for clearer roles and boundaries becomes apparent.

Developing Co-Agency

Stabilising roles and decision boundaries

Developing co-agency emerges as AI use becomes more routine and embedded. At this stage, organisations begin to formalise aspects of co-agency in response to experience. Roles are clarified, decision boundaries are articulated, and expectations become more consistent across contexts.

This phase is marked by **stabilisation**. Rather than relying solely on individual discretion, organisations establish shared understandings of when AI may be used, how outputs should be interpreted, and where responsibility lies. Governance mechanisms begin to align more closely with practice.

Key features of developing co-agency include:

- clearer articulation of decision authority and accountability,
- explicit recognition of interpretation as a role,
- documented practices and guidance informed by experience,
- and emerging escalation pathways for uncertainty or disagreement.

Stabilisation does not mean rigidity. Effective developing co-agency preserves space for judgement and adaptation while reducing ambiguity that undermines confidence or consistency. The goal is not to eliminate discretion, but to **support it**.

At this stage, organisations often confront tensions between flexibility and control. Too much formalisation can stifle learning or innovation; too little can leave individuals

exposed or practices incoherent. Co-agency development involves negotiating these tensions rather than resolving them once and for all.

Learning becomes more collective during this phase. Reflection moves beyond individual experience to include team-level or organisational review. Outcomes are discussed, not just executed. Misalignments between AI output and human judgement are examined for insight rather than blame.

Developing co-agency also requires attention to **role alignment**. Accountability must be matched with authority. Individuals expected to exercise judgement must have access to information, time, and support. Without this alignment, formal responsibility may exist without practical agency.

This phase is often where co-agency either strengthens or stalls. If stabilisation becomes overly prescriptive, co-agency may harden into procedural compliance. If stabilisation is avoided altogether, early-phase fragility persists. Capability-oriented organisations treat developing co-agency as an ongoing adjustment rather than a one-time design task.

Advanced Co-Agency

Adaptive use, reflective escalation, and learning

Advanced co-agency is characterised by **adaptability and reflection**. At this level, organisations are able to adjust co-agency arrangements in response to changing technologies, contexts, and insights. Agency is not only stabilised; it is **actively maintained**.

In advanced co-agency, roles and decision boundaries are well understood, but not fixed. Individuals know when and how to exercise judgement, and they are supported in escalating uncertainty or challenge without stigma. Escalation is treated as a normal part of responsible practice rather than a failure.

Key features of advanced co-agency include:

- confidence in exercising and challenging judgement,
- explicit mechanisms for reviewing and revising AI use,

- integration of learning into governance and practice,
- and alignment between organisational values and everyday decisions.

Advanced co-agency supports **reflective escalation**. When AI-supported decisions raise concern, individuals know where to take those concerns and what will happen next. Review processes are timely, constructive, and oriented toward improvement rather than punishment.

Learning at this stage is institutional rather than incidental. Insights from AI use inform training, governance updates, and strategic decisions. Reflection feeds back into orientation, ensuring that assumptions evolve alongside experience.

Importantly, advanced co-agency does not imply maximal automation or maximal control. It implies **appropriate agency** for context and risk. In some situations, AI may operate with minimal human intervention within well-defined bounds. In others, close human involvement may be required. Advanced co-agency allows organisations to make these distinctions deliberately rather than by default.

Advanced co-agency is also resilient. When technologies change or failures occur, organisations are able to respond without losing coherence or trust. Responsibility remains clear, learning is prioritised, and adaptation is purposeful.

Co-Agency development as continuous practice

Across these patterns, one principle remains constant: **co-agency must be renewed**. Even advanced arrangements can degrade if they are not revisited. New systems, staff turnover, external pressures, or shifting incentives can all erode agency over time.

Developing co-agency therefore involves ongoing attention to:

- how roles are understood and enacted,
- whether decision boundaries remain appropriate,
- and how learning is captured and acted upon.

By framing co-agency as a developmental capability rather than a fixed state, organisations are better equipped to navigate change without surrendering agency or accountability.

The chapters that follow examine risks and failure modes that can undermine co-agency at any stage, and explore how co-agency integrates with ethics and governance to support sustainable AI capability.

Risks and Failure Modes

Human–AI co-agency does not fail all at once. It erodes gradually, often through well-intentioned practices that go unexamined as AI use becomes normalised. The risks described in this chapter are not edge cases or rare pathologies; they are **predictable failure modes** that arise when co-agency is under-specified, unsupported, or taken for granted.

Two failure modes are particularly persistent across sectors and contexts: **automation bias and over-reliance**, and **responsibility drift and ambiguity**. They frequently co-occur and reinforce one another, creating systems that appear functional on the surface while quietly undermining judgement, accountability, and learning.

Understanding these risks is essential for maintaining healthy co-agency over time.

Automation Bias and Over-Reliance

Automation bias refers to the tendency for humans to favour, defer to, or unquestioningly accept outputs generated by automated systems, even when those outputs conflict with contextual knowledge, professional judgement, or other evidence. In AI-supported work, automation bias is rarely the result of naïveté. It is more often a product of **social, organisational, and cognitive pressures**.

AI systems often present outputs with a sense of authority: ranked lists, confident language, apparent consistency, or quantitative precision. Over time, repeated exposure to such outputs can recalibrate human expectations. What begins as assistance gradually becomes default.

Over-reliance develops when:

- AI outputs are consistently treated as “good enough,”
- human challenge is framed as inefficiency or resistance,
- or questioning AI is perceived as risky, time-consuming, or futile.

In these environments, judgement is not explicitly removed, but it is **discouraged by context**.

Automation bias is reinforced by several common conditions:

- **Time pressure:** When speed is prioritised, AI outputs offer a convenient shortcut.
- **Performance metrics:** When efficiency or throughput is measured without attention to judgement quality, reliance is rewarded.
- **Organisational signalling:** When leadership endorses AI use without equal emphasis on challenge and review, deference becomes normative.
- **Opacity:** When system behaviour is difficult to understand or interrogate, humans may defer rather than risk being wrong.

The consequences of automation bias are cumulative. Early reliance may appear harmless, but over time it reshapes behaviour. Individuals stop checking outputs as carefully. Anomalies are rationalised rather than investigated. The ability to recognise error atrophies.

Crucially, automation bias does not require fully autonomous systems. It arises readily in decision-support, generative, or advisory systems—particularly when outputs are framed as recommendations rather than suggestions.

From a co-agency perspective, automation bias represents a **failure of judgement support**, not a failure of human competence. Individuals may retain nominal authority, but lack the conditions needed to exercise it meaningfully.

Healthy co-agency counteracts automation bias by:

- legitimising scepticism and challenge,
- reinforcing the interpretive role of humans,
- and making reliance a conscious choice rather than an unexamined habit.

Where these conditions are absent, over-reliance becomes self-reinforcing. The more humans defer, the less capable they feel of intervening. The less they intervene, the more authoritative the system appears.

Responsibility Drift and Ambiguity

Responsibility drift refers to the gradual misalignment between **who appears responsible** and **who is actually able to act** in AI-supported systems. Unlike automation bias, which centres on decision acceptance, responsibility drift concerns **ownership and accountability over time**.

In early AI adoption, responsibility is often clear because use is limited and visible. As systems become embedded, responsibility can become diffuse. Multiple actors contribute to outcomes, but no single actor feels authorised or obligated to take ownership when things go wrong.

Responsibility drift commonly emerges through:

- layered workflows where AI influences decisions indirectly,
- shared or collective decision-making without clear escalation paths,
- or governance structures that assign accountability without authority.

In such contexts, individuals may be held responsible in principle but lack practical agency. They may be expected to “own” decisions shaped by AI without having had meaningful opportunity to question or alter them.

Responsibility ambiguity is particularly corrosive to learning. When outcomes are contested or harmful, organisations struggle to identify where intervention is needed.

Reviews focus on surface errors rather than structural causes. Blame may be assigned retrospectively, but little changes in practice.

Responsibility drift also interacts with automation bias. As humans defer to AI outputs, responsibility subtly shifts toward the system—at least in perception. Yet when scrutiny arises, responsibility snaps back to individuals or teams who may not have exercised genuine agency at the time.

This dynamic creates **moral and organisational tension**. Individuals feel exposed. Trust erodes. Defensive practices emerge, such as rigid adherence to AI outputs or excessive documentation designed to deflect blame rather than support learning.

From a capability perspective, responsibility drift signals a breakdown in co-agency design. Responsibility exists, but it is not supported by:

- clear decision authority,
- access to relevant information,
- or recognised escalation mechanisms.

Healthy co-agency requires responsibility to be **matched with agency**. Individuals must not only be accountable, but also empowered to act, challenge, and revise practice. Where this alignment is missing, responsibility becomes performative rather than functional.

How failure modes compound

Automation bias and responsibility drift rarely operate in isolation. Together, they create systems that appear efficient but are brittle under stress.

- Automation bias reduces human intervention.
- Reduced intervention weakens clarity about who is responsible.
- Weak responsibility discourages challenge.

- Lack of challenge reinforces reliance.

This feedback loop is difficult to detect from within, particularly when systems continue to produce acceptable outcomes most of the time. Failure modes often become visible only when stakes rise, contexts change, or errors accumulate.

Importantly, these risks are not resolved by technical fixes alone. Adding explanations, dashboards, or audit trails may help, but they do not address the underlying **capability gap**. What is needed is deliberate attention to how humans are expected and supported to exercise agency.

Recognising early warning signs

Capability-oriented organisations monitor for signals that co-agency is eroding. These signals are often behavioural rather than technical.

Common early indicators include:

- declining discussion or disagreement around AI outputs,
- increasing discomfort with overriding or questioning systems,
- unclear answers to questions about accountability,
- reliance on “the system decided” as an explanation,
- and difficulty identifying who should initiate review or change.

These signals are not evidence of failure, but opportunities for intervention. When recognised early, they can prompt reflection, clarification, and adjustment before risks become entrenched.

Reframing risk as capability maintenance

The purpose of this chapter is not to discourage AI use or to frame AI as inherently risky. Rather, it is to highlight that **risk emerges when co-agency is neglected**.

Automation bias and responsibility drift are symptoms of under-developed capability, not of bad actors or flawed intentions. Addressing them requires strengthening judgement support, clarifying responsibility, and embedding learning into practice.

The chapters that follow turn from diagnosis to integration. They explore how co-agency interacts with ethics and governance, and how it can be reinforced through decision structures rather than undermined by them. In doing so, they show how co-agency can remain robust even as AI systems become more capable and pervasive.

Integrating Co-Agency With Other Capabilities

Human–AI co-agency does not operate in isolation. While it is a distinct capability domain, it is also a **structural hinge** that connects ethical intent, governance design, and everyday practice. When co-agency is weak or under-specified, other capabilities may exist in name but fail in execution. When co-agency is strong, ethical commitments become actionable and governance structures become enabling rather than obstructive.

This chapter focuses on two critical integrations: co-agency with **ethics and accountability**, and co-agency with **governance and decision structures**. Together, these relationships determine whether AI-supported systems remain trustworthy, adaptable, and human-centred over time.

Ethics and Accountability

Ethical principles are often articulated at a high level: fairness, transparency, responsibility, harm reduction, respect for persons. These principles are widely endorsed, yet organisations consistently struggle to enact them in practice. One reason for this gap is that ethics is frequently treated as a set of constraints or values rather than as a **capability that depends on agency**.

Co-agency is the mechanism through which ethics becomes operational.

Ethical accountability requires that someone can:

- recognise when a value is at stake,
- interpret how that value applies in context,
- and act on that interpretation in a timely and legitimate way.

Without clear co-agency, ethical responsibility becomes diffuse. Individuals may recognise ethical concerns but feel unable to intervene. Teams may assume that responsibility sits elsewhere. Governance bodies may articulate values without visibility into how decisions are actually made.

Co-agency addresses this gap by clarifying who is expected to notice, who is authorised to decide, and who is accountable for outcomes in AI-supported work.

Ethics as situated judgement

Ethical issues in AI rarely present themselves as obvious rule violations. More often, they appear as trade-offs, tensions, or uncertainties: efficiency versus fairness, consistency versus contextual sensitivity, innovation versus risk. These situations cannot be resolved by principles alone. They require judgement.

Co-agency provides the conditions under which ethical judgement can be exercised:

- humans retain interpretive authority,
- AI outputs are treated as inputs rather than verdicts,

- and escalation pathways exist when ethical uncertainty arises.

Where co-agency is weak, ethical reasoning is displaced. Individuals may defer to AI outputs even when they conflict with values, or rely on policy language to avoid making difficult judgements. Ethics becomes retrospective—invoked after harm occurs—rather than prospective and embedded in practice.

Accountability without agency is performative

Many organisations assign ethical accountability without ensuring corresponding agency. Individuals are told they are “responsible” for AI-influenced decisions, yet lack meaningful control over system design, output interpretation, or escalation processes.

This mismatch creates performative accountability. Responsibility exists on paper, but not in practice. When issues arise, individuals may be blamed for outcomes they could not realistically influence.

Healthy co-agency aligns accountability with authority. It ensures that those who are expected to uphold ethical standards:

- have access to relevant information,
- are empowered to question or override AI outputs,
- and are supported when ethical concerns require delay, review, or refusal.

This alignment is essential not only for fairness, but for sustainability. Systems that repeatedly place individuals in untenable positions erode trust and invite defensive behaviour.

Ethics as an ongoing capability

Ethics is not resolved at design time. Values must be interpreted repeatedly as contexts change, uses expand, and impacts become visible. Co-agency supports this ongoing ethical capability by embedding reflection and learning into decision-making processes.

When co-agency is integrated with ethics:

- ethical concerns are surfaced earlier,
- responsibility is shared rather than displaced,
- and learning feeds back into future practice.

In this sense, co-agency does not constrain ethical ambition; it enables it. Without co-agency, ethics remains aspirational. With co-agency, ethics becomes lived practice.

Governance and Decision Structures

Governance is often misunderstood as a set of controls imposed on practice. In reality, governance shapes how agency is distributed, constrained, and supported. Governance that ignores co-agency either centralises authority excessively or disperses responsibility so widely that no one can act.

Co-agency and governance are therefore inseparable.

Governance answers questions such as:

- Who is authorised to decide?
- Under what conditions?
- With what oversight?
- And how are decisions reviewed, challenged, or revised?

Co-agency determines whether those answers are meaningful in practice.

Governance as an enabler of agency

Effective governance does not remove judgement; it **structures it**. It clarifies boundaries so that individuals know when they can act autonomously, when consultation is required, and when escalation is expected.

When governance is aligned with co-agency:

- decision rights are explicit,
- accountability pathways are visible,
- and uncertainty is treated as a signal for review rather than failure.

In contrast, governance that is detached from co-agency often produces one of two outcomes:

- excessive rigidity, where individuals defer all decisions upward and innovation stalls;
- or informal workarounds, where governance is bypassed to “get things done.”

Both outcomes undermine capability.

Decision structures and escalation

AI-supported decisions often sit at the intersection of multiple domains: technical feasibility, ethical acceptability, operational impact, and reputational risk. No single individual or role can reasonably hold all relevant expertise.

Co-agency requires decision structures that support **collective judgement without diffusing responsibility**. This includes:

- clear escalation routes when AI outputs are contested,
- mechanisms for pausing or revisiting decisions,
- and defined roles for review and oversight.

Without these structures, individuals may feel compelled to accept AI outputs even when uncertain, simply because no alternative pathway exists.

Escalation should not be framed as failure. In a capability-oriented system, escalation is a sign that co-agency is functioning: uncertainty is recognised, responsibility is taken seriously, and decisions are revisited rather than rushed.

Governance across the lifecycle

Governance is not static. As AI use matures, decision structures must adapt. Early experimentation may require lighter governance and closer human oversight. As practices stabilise, governance can shift toward consistency and accountability. When contexts change, governance must be revisited.

Co-agency provides continuity across these phases. It ensures that governance evolves in response to lived practice rather than being frozen at an early moment of uncertainty or enthusiasm.

Capability-oriented governance therefore:

- listens to signals from practice,
- adjusts decision structures accordingly,
- and maintains alignment between responsibility and authority over time.

Avoiding governance drift

Just as responsibility can drift, so can governance. Policies may remain formally intact while actual practice diverges. New tools may be adopted without revisiting existing decision structures. Informal norms may replace documented processes.

Strong co-agency counteracts governance drift by keeping decision-making visible and discussable. When humans remain active agents rather than passive operators, misalignments between governance and practice surface earlier and can be addressed deliberately.

Co-Agency as connective tissue

Ethics without agency is aspiration. Governance without agency is bureaucracy. Co-agency is what allows both to function in practice.

By integrating co-agency with ethics and governance, organisations move beyond compliance and control toward **capability and confidence**. Decisions remain human-centred even as AI systems become more capable. Responsibility is exercised rather than assigned. Learning is embedded rather than episodic.

This integration is what allows AI capability to scale without fragmenting.

The following chapter turns from integration to application, examining how co-agency is enacted differently across education, research, and organisational contexts—while remaining grounded in the same underlying principles.

Applying Co-Agency in Context

Human–AI co-agency is not enacted in the abstract. It takes shape through concrete decisions, institutional norms, and professional responsibilities that vary across contexts. While the underlying principles of co-agency remain consistent—clarity of roles, retained human judgement, accountable decision-making—the way these principles are expressed differs significantly between education, research, and organisational or public service settings.

This section explores how co-agency operates in each of these contexts, highlighting both its distinctive challenges and its unifying function. The aim is not to prescribe uniform solutions, but to show how co-agency can be interpreted and enacted in ways that respect contextual purpose while maintaining capability coherence.

Education

Teaching, Assessment, and Student Agency

In education, co-agency is inseparable from questions of learning, authorship, and trust. Educational institutions are not only users of AI; they are formative environments in which learners develop norms about knowledge, judgement, and responsibility. As a result, failures of co-agency in education have consequences that extend beyond immediate outcomes, shaping how future professionals understand their own agency in relation to technology.

Teaching and pedagogical judgement

In teaching contexts, AI systems are increasingly used to support lesson planning, content generation, feedback design, and learning analytics. Co-agency here requires clarity about where AI supports pedagogical judgement and where it must not substitute for it.

Educators retain responsibility for:

- defining learning outcomes,
- interpreting student needs,
- and exercising professional judgement about what constitutes meaningful learning.

When AI tools are treated as authoritative sources rather than supports, pedagogical agency erodes. Teaching risks becoming content delivery rather than relational, interpretive practice. Co-agency preserves the educator's role as designer and judge, with AI operating as an assistive resource rather than a pedagogical actor.

Assessment and academic judgement

Assessment is one of the most sensitive sites of co-agency in education. Decisions about grading, progression, and feedback carry significant consequences for learners. Introducing AI into assessment workflows without clear co-agency risks undermining both fairness and legitimacy.

Healthy co-agency in assessment requires:

- explicit boundaries on where AI may assist (e.g. rubric alignment, feedback drafting),
- retained human authority over evaluative judgement,
- and transparent communication with students about AI's role.

Over-automation in assessment can produce the appearance of objectivity while concealing interpretive choices embedded in systems. Conversely, blanket prohibition often drives AI use underground, eroding trust. Co-agency enables a middle path: AI supports assessment work without displacing accountability.

Student agency and learning integrity

Co-agency in education is not limited to staff. Students are also agents interacting with AI systems, often with fewer resources and less guidance. Institutions that focus solely on staff use while ignoring student agency create asymmetries that undermine integrity.

Supporting student co-agency involves:

- helping learners understand when AI can support learning and when it undermines it,
- modelling reflective use rather than surveillance-based control,
- and recognising that students' agency develops through guided practice, not prohibition.

When students are treated only as compliance subjects, capability development is stunted. When they are treated as developing agents, co-agency becomes part of professional formation rather than a disciplinary issue.

Research

Authorship, Interpretation, and Methodological Integrity

In research contexts, co-agency is tightly coupled to epistemic responsibility. Research is not only about producing outputs, but about justifying claims, interpreting evidence, and contributing to shared knowledge. AI systems can influence each of these stages, often subtly.

Authorship and contribution

One of the most visible co-agency challenges in research concerns authorship. AI tools can generate text, suggest structures, summarise literature, and propose interpretations. Without clear co-agency, responsibility for scholarly claims becomes blurred.

Co-agency requires that:

- human researchers remain accountable for intellectual contributions,
- AI use is disclosed where relevant,
- and authorship is not conflated with output production.

The core question is not whether AI was used, but **who is responsible for the reasoning and claims presented**. Co-agency preserves the integrity of authorship by ensuring that AI assistance does not obscure responsibility.

Interpretation and epistemic judgement

Research depends on interpretation: deciding what findings mean, how evidence supports claims, and where uncertainty remains. AI systems may assist with pattern detection or synthesis, but they do not understand context, theory, or consequence.

Failures of co-agency in research often occur when AI outputs are treated as interpretations rather than inputs. This can lead to overconfidence, methodological shortcuts, or unexamined bias.

Strong co-agency ensures that:

- AI-supported analysis is subject to critical scrutiny,
- interpretive authority remains human,
- and uncertainty is acknowledged rather than smoothed over.

This is particularly important in interdisciplinary research, where assumptions embedded in AI systems may not align with domain-specific epistemologies.

Methodological integrity over time

AI tools evolve rapidly, and methods that appear robust today may be problematic tomorrow. Co-agency supports methodological integrity by embedding reflection and review into research practice.

Rather than fixing rules about permissible tools, co-agency focuses on:

- ongoing evaluation of how AI affects research design,
- shared learning across research groups,
- and adaptive governance that responds to emerging norms.

In this way, co-agency supports innovation without compromising rigour.

Organisations and Public Service

Operational Decisions and Leadership Responsibility

In organisational and public service contexts, co-agency is most directly linked to accountability and legitimacy. AI-supported decisions can affect employment, access to services, resource allocation, and public trust. Here, the stakes of agency misalignment are particularly high.

Operational decision-making

AI is increasingly used to support operational decisions: prioritisation, risk assessment, forecasting, and case handling. Co-agency requires that these systems do not become de facto decision-makers.

Healthy co-agency ensures that:

- humans understand how AI outputs are generated,
- decision boundaries are clearly defined,
- and overrides are legitimate and supported.

When AI outputs are treated as defaults rather than recommendations, responsibility quietly shifts from people to systems. This shift is rarely explicit, but it has profound implications for accountability.

Leadership and responsibility

Leadership plays a central role in establishing co-agency. Leaders set expectations about whether AI use is reflective or performative, whether questioning outputs is encouraged or discouraged, and whether responsibility is owned or displaced.

Capability-oriented leadership:

- models critical engagement with AI,
- supports staff who raise concerns or escalate uncertainty,

- and resists narratives of inevitability or replacement.

Where leadership abdicates agency—by deferring to vendors, tools, or external pressure—co-agency collapses throughout the organisation.

Public trust and legitimacy

In public service contexts, legitimacy depends on visible accountability. Citizens must be able to understand who is responsible for decisions that affect them, even when AI systems are involved.

Co-agency supports legitimacy by:

- keeping decision authority human and identifiable,
- enabling explanation and contestation,
- and embedding learning from impact and feedback.

Without co-agency, AI use may appear efficient but opaque, eroding trust even when outcomes are technically sound.

A shared capability, differently expressed

Across education, research, and organisational contexts, the expression of co-agency varies. What remains constant is its function: **to keep humans accountable, judgement-bearing, and capable of learning in AI-supported systems.**

Co-agency is not a constraint on innovation. It is what allows innovation to persist without eroding responsibility. It is not an abstract ethical ideal, but a practical capability that shapes how decisions are made, justified, and revised over time.

The final chapter turns inward, focusing on reflection and renewal—how co-agency itself must be revisited, evaluated, and sustained as technologies and contexts evolve.

Reflection and Renewal

Human–AI co-agency cannot be established once and then assumed to persist. It is shaped continuously by changes in technology, organisational context, professional norms, and social expectations. Systems that appear well-designed at one moment may quietly drift at another. Practices that feel responsible in early experimentation may become risky once scaled. Roles that were once clear may blur as tools become more capable and more embedded.

For these reasons, reflection and renewal are not optional additions to co-agency. They are the mechanisms through which co-agency is sustained.

Reflection allows individuals and organisations to notice how agency is actually operating in practice, rather than how it is assumed to operate. Renewal translates that noticing into adjustment: revisiting boundaries, updating norms, and re-asserting human responsibility where it has weakened or become unclear.

This chapter focuses on two complementary aspects of that work: reflective prompts that support inquiry at individual and collective levels, and the conditions required to sustain healthy co-agency over time.

Reflective Prompts for Individuals and Teams

Reflection in the context of co-agency is not about self-assessment or compliance. It is about sense-making. It creates space to ask whether responsibility, judgement, and authority are aligned in lived practice, not just in policy or design.

The prompts below are not intended to be answered definitively. Their value lies in being revisited periodically, particularly after moments of change, uncertainty, or tension.

Reflection on judgement and decision-making

- When AI systems contribute to decisions in my work, how do I understand my own role? Do I feel like the decision-maker, the reviewer, or the executor of an output?
- Are there situations where I defer to AI outputs more than I am comfortable admitting? What pressures or assumptions make that deferral feel necessary?

- When I disagree with an AI-supported recommendation, do I feel authorised and supported to challenge it?

These questions help surface whether judgement is being actively exercised or quietly displaced. Discomfort, hesitation, or ambiguity are signals worth attending to rather than suppressing.

Reflection on responsibility and accountability

- If an AI-supported decision led to harm or scrutiny, would it be clear who is accountable?
- Do accountability structures in my context match actual decision authority, or do they assign responsibility without agency?
- Where do I rely on policy, process, or technology to carry responsibility that I am unwilling or unable to hold myself?

These prompts reveal whether responsibility is genuinely owned or rhetorically assigned. Persistent misalignment between accountability and authority is one of the strongest indicators of fragile co-agency.

Reflection on learning and adaptation

- How do we currently learn from AI use in our team or organisation?
- Do we treat unexpected outcomes as failures to hide, or as signals to examine?
- When practices change, do we understand why, or do changes happen by drift?

Reflection without learning leads to stagnation. Learning without reflection leads to repetition. Co-agency depends on both.

Reflection on culture and norms

- What behaviours are implicitly rewarded or discouraged in relation to AI use?
- Is questioning AI outputs seen as diligence or resistance?
- Do we make space for uncertainty, or do we expect confidence even when understanding is partial?

Cultural signals often shape agency more powerfully than formal guidance. These prompts help surface whether the environment supports thoughtful engagement or quiet compliance.

Used collectively, these reflections support shared understanding. Used individually, they help professionals retain agency even within complex systems. In both cases, reflection is a practice of responsibility, not introspection for its own sake.

Sustaining Healthy Co-Agency

Reflection alone does not sustain co-agency. Renewal requires action: adjustments to structures, norms, and expectations that keep agency aligned with responsibility over time.

Healthy co-agency is not static. It must be actively maintained against several predictable pressures.

Pressure toward automation and convenience

As AI systems become more capable and more integrated, the temptation to offload judgement increases. What begins as assistance can gradually become default authority, particularly when systems are reliable most of the time.

Sustaining co-agency requires deliberate friction:

- moments where humans must interpret rather than accept,

- expectations that outputs are questioned rather than consumed,
- and visible endorsement of human judgement, even when it slows processes.

Convenience is not neutral. Left unchecked, it erodes agency.

Pressure toward diffusion of responsibility

In complex systems, responsibility often becomes diluted. Multiple actors touch a decision, but no one feels fully accountable. AI systems can intensify this diffusion by acting as intermediaries that obscure where choices are made.

Healthy co-agency resists diffusion by:

- clarifying who decides, who reviews, and who learns,
- making responsibility explicit rather than assumed,
- and revisiting these clarifications as systems evolve.

Responsibility must be repeatedly re-claimed, not assumed to persist.

Pressure toward performative compliance

As governance and ethical expectations increase, organisations may respond by producing policies, statements, or assurances that are weakly connected to practice. Co-agency collapses when responsibility becomes performative rather than enacted.

Renewal involves reconnecting formal structures to lived decisions:

- testing whether governance supports or constrains judgement,
- checking whether ethical commitments shape real choices,
- and adjusting where gaps appear.

Capability is revealed in practice, not documentation.

Supporting renewal through structure

Sustaining co-agency benefits from structural support:

- regular moments of review built into workflows,
- forums where AI use can be discussed without defensiveness,
- and leadership behaviours that legitimise questioning and escalation.

Without these supports, reflection becomes individualised and fragile. With them, co-agency becomes collective and resilient.

Leadership and renewal

Leaders play a decisive role in sustaining co-agency. Not by issuing directives, but by modelling how responsibility is held.

Leadership that supports co-agency:

- acknowledges uncertainty,
- treats reflection as strength rather than hesitation,
- and demonstrates willingness to revise decisions.

Where leaders abdicate agency—by deferring to tools, vendors, or inevitability narratives—co-agency weakens throughout the system.

Co-agency as a commitment, not a state

Ultimately, sustaining healthy co-agency requires a shift in how success is understood. Success is not the absence of friction, nor the seamless integration of AI into workflows. Success is the ongoing capacity to notice, question, and adapt.

Co-agency is not something an organisation *has*. It is something an organisation *does*, repeatedly.

It is renewed whenever:

- a human pauses to interrogate an output,
- a team revisits a boundary that no longer holds,
- or an institution chooses accountability over convenience.

This book has argued that co-agency is central to AI capability because it keeps humans answerable for decisions in environments shaped by increasingly powerful systems. Reflection and renewal are what keep that answerability alive.

There is no final configuration in which co-agency is complete. There is only continued attention, exercised judgement, and the willingness to adapt responsibly as conditions change.

That ongoing work is not a limitation. It is the condition under which AI remains a tool for human purposes rather than a force that quietly reshapes them.

Sustaining Co-Agency Over Time

Human–AI co-agency is not something an organisation can design once and then assume will endure. It is shaped continuously by technological change, organisational pressure, professional norms, and cultural signals about authority and responsibility. Even well-designed co-agency arrangements can erode quietly if they are not revisited and reinforced.

Throughout this book, co-agency has been framed not as a technical safeguard, but as a practical capability: one that determines whether AI augments human judgement or gradually displaces it. The distinction matters. When co-agency is healthy, responsibility is visible, judgement is exercised confidently, and learning is possible. When it weakens, accountability becomes diffuse and decision-making loses legitimacy.

Co-agency is sustained through attention rather than control. It is renewed whenever individuals feel authorised to question AI outputs, when escalation is treated as responsible practice rather than failure, and when organisations choose clarity over convenience. These moments are rarely dramatic, but they are decisive.

There is no final or optimal configuration of co-agency. What counts as appropriate human involvement will vary by context, risk, and maturity, and will need to be recalibrated as systems and practices evolve. The goal is not to minimise automation or maximise oversight, but to ensure that agency remains aligned with responsibility.

This book should therefore be read as a guide for ongoing work rather than a set of rules. Its purpose is to help readers notice when agency is drifting, articulate where responsibility truly lies, and intervene deliberately before erosion becomes normalised.

Co-agency is not something an organisation has. It is something an organisation does—again and again—as AI becomes more capable and more persuasive. Sustaining that practice is what keeps AI accountable to human purposes rather than quietly reshaping them.

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